The Making of a Grass-Root Plot Plan

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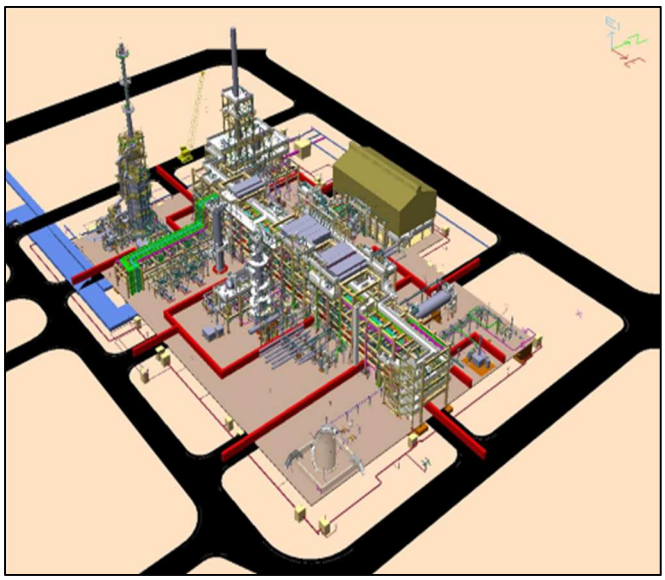
Overview:



Development of plot plan by continual refinement of equipment layout is of critical importance in a project. Inclusion of contributions from diverse functional experts through model reviews and comments in the drawing makes the process a collaborative effort at the conceptual stage. The urgency to generate work front to downstream functional groups at an early stage adds to its complexity and makes this process more challenging.

This article shares the experience of our own in-house engineering wing of Reliance: REG-JEC, while developing their first ISBL unit Plot plan of J3 as a DEC. Experience of grass root unit (DTA SCANfiner - Green field) is shared for the benefit of all stakeholders of One –Reliance. This article unfolds the evolution of Scanfiner plot plan at various stages and provides information to contribute and manage the work process of developing future plot plans in a new transformed way.

Keywords: Equipment Layout, Conceptual layout, Design Development, Value Engineering, Change Management, Work process Improvement

Abbreviations: DTA: Domestic Tariff Area; REG: Reliance Engineering Group; JEC: Jamnagar Engineering Centre; DEC: Detail Engineering Contractor; CAR: Corrective Action Report; PAR: Preventive Action Report; PMT: Project Management Team;

Conceptual Stage:

Major challenge at this stage was to overcome the inertia and start developing conceptual layout especially when many critical inputs were awaited. Instead of waiting for the location coordinates and size of Scanfiner unit in the site plan, REG spearheaded the project by developing various layout options with assumed size of the plot and offsite rack configuration. The site selection of the unit was done considering the safety aspects like all around road, access to unit from offsites existing rack, etc. The encumbrances in the plot area like existing well, site offices etc. were brought to notice for finalizing the location and pipe rack entry of the unit [1].

A study report was generated for finalizing the rack design at the early stage of the project. Initial equipment layout was developed based on preliminary equipment list, PFDs by the licensor and agreed decisions from projects on unit location, rack design, pipe rack battery limit, etc. Critical equipment size like for heater and compressor were assumed based on available data in datasheet and historical data.

The initial equipment layout was similar to existing SEZ Scanfiner unit (Fig 1) with straight pipe rack with 23 bays. The critical issue with this layout was many critical piping of the unit between reactors & exchangers as well as piping between heater and its charge pump was iterating in the main pipe rack. Apart from the increased length of alloy steel piping the rack design was getting complicated due to the loads of critical piping of high temperature.

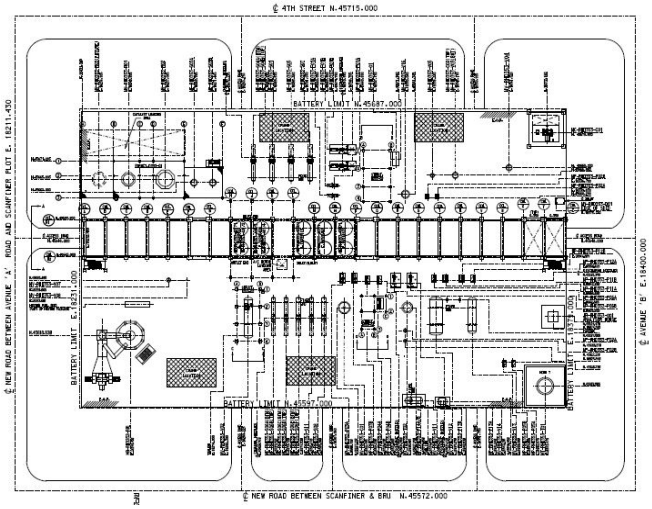


Fig 1

The solution was arrived by converting the conventional linear layout to T type (Fig.2).[4] Apart from saving of 3 bays and 20% reduction in length of critical piping in main pipe rack all the major critical high temperature lines were relocated from main pipe rack to spur rack. This layout helped in early release of main pipe rack as construction front. When this plot plan was given to Licensor for review it received appreciation by specialists in this field. [5] Their feedback were: "The plot incorporates all the fundamentals of a good layout namely safety, accessibility, constructability, maintenance, economy and operability."

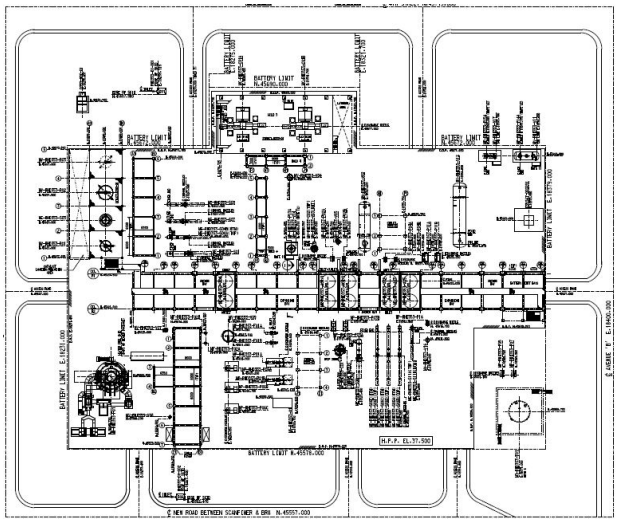


Fig 2

The resulted in the additional cascading benefit of saving extra staircase. There was a requirement to provide staircase for reactor structure and both ends of pipe rack. With this new layout platform connectivity was possible between reactor and fin fan cooler platform which provided the opportunity to have common staircase between reactor structure and pipe rack. This configuration of platform was highly appreciated by O&M during 30% model review.

All the functional groups like construction, O&M, PMT, HSE and all disciplines of REG actively participated in the commenting process of the plot plan. All the comments from the diverse functional group further refined the plot plan prior to IFD.

Key Challenges:

a) Site selection

b) Unit plot size

c) Location of ISBL unit in site plan

d) OSBL Rack configuration to finalize pipe rack B/L of ISBL unit

e) ISBL Pipe Rack design finalization

f) Early release of pipe rack drawings as construction front

Implemented Solutions:

a) Safety aspect like all around road, existing fire water network availability, influence of existing site offices etc. were considered while selecting location of the unit at site.

b) Study site plan with various layout options for key challenges as per a), b), c) and d). [1]

c) Study report of rack design options for key challenge e) [2]

d) Value engineered layout option with T type rack configuration against key challenge f).

Benefits:

a) Early finalization of unit location, size of plot and pipe rack B/L entry which was later updated in the site plan drawings.[3]

b) Early finalization of ISBL rack design option.

c) Early release of main pipe rack [6]

Design Development Stage:

Major challenge at this stage was to comply with the robust procedures of REG quality management system. Risk assessment was done and risk and action plan was informed to project engineering manager. The work proceeded after the approval of these waivers by PEM. [7] Risk assessment approach, which is currently prevalent in Reliance and new ISO 9001:2015 standard was utilized in this project.

Design development progressed with 30%, 60%, model reviews followed by series of internal and PMT audits. The snapshot of design development is documented by way of model review reports in REIMS. [8]

Exemplary model reviews were conducted. Main highlights of model reviews were

a) All delegates were given the evacuation route drawing of the conference room on their desk with identification of exit doors for escape in case of emergency.
b) Intelligent CAD indexing [9] was explained and each CAD areas were logically unveiled in a proper sequence which was appreciated by all.
c) Opportunities were given to young engineers to present their areas as Area leads. This evoked team spirit and ownership mindset which was very much essential for the project.
d) Highly interactive sessions where listening to all functional group was given prime importance. Plethora of design refinement issues were discussed and agreed points were tagged for documentation which can be issued for future reference.[8]
e) Heater area was outsourced to vendor for modeling using work share process and they seamlessly participated in model reviews. This outsourcing experience is first of its kind in REG which provided exposure to working in collaboration with external group.
f) Licensor reviewed the model system wise and provided rating of 9/10 during additional 60% model review.

FIG 3

FIG 4

During the regular quality audits no CARS and PARS were observed. PMT auditors highlighted the strengths of engineering which were agreed to be emulated to all DEC in future opportunities. Major appreciated points during PMT audits were: [11]

a) Equipment list and corresponding grid is shown on plot plan, which is an improvement to be followed by other DECs. (Piping - Plot Plan) [10]

b) Future space provision is clearing earmarked by a separate envelope. (Piping - Model Review)

c) JEC-DCC has initiated efforts towards Paperless DCC, which will reduce papers resource and time (DCC)

Feedback from project manager on PMT audit:

"We too had audit last Monday and the auditors stated that JEC is the best they have audited so far. This is another excellent remark which we have to behold" as told by project manager.

Key Challenges:

a) Non availability of minimum inputs for plot plan IFA stage as per procedure

b) Exclusions and assumptions in 30% & 60% model reviews [8]

c) Integration of vendor efforts in heater area in 3D model environment

Implemented Solutions:

a) Deviations were approved by informing the risk to PEM. Risk assessment was used by REG [7]

b) Conducted model reviews by publishing the exclusions and assumptions to all stakeholders. [8]

c) Work share established for heater area with vendor and 3D model set up and synchronization enabled seamless integrated work with external vendor in one 3D model environment.

Benefits:

a) Project progress with a risk assessment approach

b) Design development progress by having early model reviews

c) Rework of vendor modeled efforts saved and unnecessary vendor interfaces were reduced resulting in substantial saving in schedule.

Change Management:

Change is the only constant in any Reliance project. It brings new challenges for which one has to orient oneself and raise to the occasion to meet the objectives. Change in this project was evident from the beginning and brought new challenges and insight in the entire journey of design development. Few major changes and its management is mentioned below:

A. Change in compressor type after 30% model review: This single change was having a potential to disrupt conceptual layout from roots. Exhaustive study was done from the entire list of existing compressor available in Jamnagar Refinery to propose a new modification with minimum impact to the original configuration. Comparison of compressor area of fig 1 and fig 2 shall give the insight of the change.

B. Shift in reactor foundation coordinates: This change demonstrated the values of excellence and team work. REG took a positive approach and explored all the possible options to support our construction team. Worked together with them to get the as-built coordinates and re-engineered the entire critical piping connecting to reactor structure and spur rack. Stress team and CSA group showed exemplary performance in a short duration to redesign the spur rack and associated piping. This event developed an reputation among other functional groups to have an in- house engineering team to rely upon.

C. Shift in compressor foundation coordinates: This change even though was very critical from engineering aspect but was addressed in a seemingly effortless way. The experience of previous re - engineering gave sufficient expertise to handle this change. All the critical compressor critical line were re-analyzed and model review was done with all stakeholders to arrive at the solution. Complete support was provided the construction team and the re-work avoided.

Key Challenges:

a) Change from one centrifugal compressor to two reciprocating ones after 30% model review and conceptual stage of plot plan development.

b) Shift in Reactor foundation coordinates

c) Shift in Compressor foundation coordinates

Implemented Solutions:

a) Accommodated this change in existing location with minimum modification with marginal increase in overall plot size. Used the historical data of existing compressors to finalize the space and ensured that vendor provides the configuration within the available dimensions.

b) As-built coordinates for reactor were obtained from construction team and complete re- engineering was done in record time. Excellent team work observed between CSA and Stress piping team.

c) As-built coordinates for compressor were obtained from construction team and complete re-engineering was done.

Benefits:

a) Saved approx. 6 months of engineering efforts by coming out with a modification with minimum impact on the conceptual layout.

b) Saved huge construction rework which had a potential to deeply impact the project schedule and cost

c) Saved major construction rework and saved substantial project schedule and cost

Lessons Learnt and creation of new system:

All lesson learnt from the project is entered in lesson learnt portal. There are total 37 lessons learnt. Few of the pertaining to plot plan development are:

a) Check HPP elevation w.r.t. all around road elevation to comply with IS 5572 / Area classification requirement.

b) Number of plot plan IFC issuance in revisions shall be finalized upfront in a project. Preferably provisions for five revisions shall be kept.

Based on the above learnings and experience of J3 project plot plan checklist was revised and special focus was emphasized on statutory checks and Safety checks along with other equipment layout related checks.[12]

Tabular Representation:

Sr. No.	Challenges	Approach	Cost and Schedule Impact
1.	Site selection, Conceptual layout finalization	Safety, Layout options, Study Reports, Value Engineered layout option	Early kick start of project. Saving of project schedule.
2.	Exclusions and assumptions in 30% & 60% model reviews	Conducted model reviews by publishing the exclusions and assumptions to all stakeholders	Design development progress by having early model reviews
3.	Integration of vendor efforts in heater area in 3D model environment	Work share established for heater area with vendor and 3D model set up and synchronization enabled seamless integrated work with external vendor in single 3D model environment.	Rework of vendor modeled efforts saved and unnecessary vendor interfaces were reduced resulting in substantial saving in schedule
4.	Change from one centrifugal compressor to two reciprocating ones after 30% model review and conceptual stage of plot plan development.	Accommodated this change in existing location with minimum modification with marginal increase in overall plot size. Used the historical data of existing compressors to finalize the space and ensured that vendor provides the configuration within the available dimensions.	Saved approx. 6 months of engineering efforts by coming out with a modification with minimum impact on the conceptual layout.

5.	Shift in Reactor and Compressor foundation coordinates	As-built coordinates were obtained from construction team and complete re- engineering was done.	Saved huge construction rework which had a potential to deeply impact the project schedule and cost

Conclusion

The journey of developing the first ISBL grass root plot plan of Reliance J3 plant has given substantial new challenges which was not envisaged earlier. With this new learnings and insights REG has gained greater confidence in facing such challenges.

This high level of design refinement in plot plan was possible with the active participation of all stakeholders in model reviews and their comments in the plot plan. We acknowledge this true demonstration of One Team Value.

As saying of old proverb goes:

"Knowledge does not diminish by dissemination but on the contrary it grows."

The recently launched Digital J3 enterprise shall provide a perfect platform for the dissemination and augmentation of the knowledge to a higher level.

References:

[1] 10074-P65-RHD353-003 dated 19.12.12; 10074-P65-RHD353-004 dated 19.12.12; 10074-P65-RHD353-005 dated 28.12.12; 10074-P65-RHD353-006 dated 14.02.13

[2] 10074-P65-RHD353-012 dated 02.02.13

[3] 10015-P15-JG0000-1000; 10015-P65-JGP000-154; 10015-P70-JG0000-1000;

[4] E-mail communication from REG proposing value engineered layout option

[5] E-mail communication from Licensor as feedback to conceptual layout.

[6] 10074-P50-RHD353-004 Rev 0 dated 19.07.2013

[7] 10074-PDV-RHD353-0003 dated 4th June 2013

[8] Model review reports: 10074-P73-RHD353-001 to 005

[9] Intelligent indexing is key to 3D CAD plant modeling by Dr. S. Saha and A. Nair; published in Hydrocarbon Processing Dec'2012

[10] Suggestion no: JEC/PP/10/2012/0027

[11] Email communication from PMT on final audit report: Fuel Refinery, Gasoline Upgrade, AUD-14-105

[12] Revised Plot Plan Checklist JEC-CL-PP-7602 Rev 0